In [1]:
```
import numpy as np
import pandas as pd
```

In [2]:
```
df=pd.read_csv("student_performance.csv")
df
```

Out[2]:

	student_id	name	course	python_marks	numpy_marks	pandas_marks
0	101	Arun	Python	85.0	78.0	92
1	102	Priya	Python	91.0	88.0	95
2	103	Karthik	NumPy	76.0	82.0	79
3	104	Divya	Pandas	89.0	94.0	90
4	105	Rahul	Python	67.0	73.0	70
5	106	Meena	NumPy	95.0	91.0	97
6	107	Vijay	Pandas	72.0	68.0	75
7	108	Ananya	Python	88.0	85.0	91
8	109	Suresh	NumPy	NaN	77.0	81
9	110	Lakshmi	Pandas	93.0	96.0	94
10	111	Naveen	Python	58.0	64.0	61
11	112	Keerthi	NumPy	81.0	86.0	84
12	113	Manoj	Pandas	74.0	NaN	78
13	114	Sneha	Python	97.0	93.0	98
14	115	Ravi	NumPy	69.0	72.0	74

In [3]:
```
df.head()
```

Out[3]:

	student_id	name	course	python_marks	numpy_marks	pandas_marks
0	101	Arun	Python	85.0	78.0	92
1	102	Priya	Python	91.0	88.0	95
2	103	Karthik	NumPy	76.0	82.0	79
3	104	Divya	Pandas	89.0	94.0	90
4	105	Rahul	Python	67.0	73.0	70

In [4]:
```
df.tail()
```

Out[4]:

	student_id	name	course	python_marks	numpy_marks	pandas_marks
10	111	Naveen	Python	58.0	64.0	61
11	112	Keerthi	NumPy	81.0	86.0	84
12	113	Manoj	Pandas	74.0	NaN	78
13	114	Sneha	Python	97.0	93.0	98
14	115	Ravi	NumPy	69.0	72.0	74

In [9]:
```
df.columns
```

Out[9]:
```
Index(['student_id', 'name', 'course', 'python_marks', 'numpy_marks',
       'pandas_marks'],
      dtype='str')
```

In [10]:
```
df.info()
```

```
<class 'pandas.DataFrame'>
RangeIndex: 15 entries, 0 to 14
Data columns (total 6 columns):
 #   Column         Non-Null Count  Dtype
---  ------         --------------  -----
 0   student_id     15 non-null     int64
 1   name           15 non-null     str
 2   course         15 non-null     str
 3   python_marks   14 non-null     float64
 4   numpy_marks    14 non-null     float64
 5   pandas_marks   15 non-null     int64
dtypes: float64(2), int64(2), str(2)
memory usage: 852.0 bytes
```

In [11]:
```
df.describe()
```

Out[11]:

	student_id	python_marks	numpy_marks	pandas_marks
count	15.000000	14.000000	14.000000	15.000000
mean	108.000000	81.071429	81.928571	83.933333
std	4.472136	11.925824	10.171334	11.054842
min	101.000000	58.000000	64.000000	61.000000
25%	104.500000	72.500000	74.000000	76.500000
50%	108.000000	83.000000	83.500000	84.000000
75%	111.500000	90.500000	90.250000	93.000000
max	115.000000	97.000000	96.000000	98.000000

In [12]:
```
df.isnull()
```

Out[12]:

	student_id	name	course	python_marks	numpy_marks	pandas_marks
0	False	False	False	False	False	False
1	False	False	False	False	False	False
2	False	False	False	False	False	False
3	False	False	False	False	False	False
4	False	False	False	False	False	False
5	False	False	False	False	False	False
6	False	False	False	False	False	False
7	False	False	False	False	False	False
8	False	False	False	True	False	False
9	False	False	False	False	False	False
10	False	False	False	False	False	False
11	False	False	False	False	False	False
12	False	False	False	False	True	False
13	False	False	False	False	False	False
14	False	False	False	False	False	False

In [13]:
```
df.isnull().sum()
```

Out[13]:
```
student_id      0
name            0
course          0
python_marks    1
numpy_marks     1
pandas_marks    0
dtype: int64
```

In [14]:
```
df.duplicated()
```

Out[14]:
```
0     False
1     False
2     False
3     False
4     False
5     False
6     False
7     False
8     False
9     False
10    False
11    False
12    False
13    False
14    False
dtype: bool
```

In [15]:
```
df
```

Out[15]:

	student_id	name	course	python_marks	numpy_marks	pandas_marks
0	101	Arun	Python	85.0	78.0	92
1	102	Priya	Python	91.0	88.0	95
2	103	Karthik	NumPy	76.0	82.0	79
3	104	Divya	Pandas	89.0	94.0	90
4	105	Rahul	Python	67.0	73.0	70
5	106	Meena	NumPy	95.0	91.0	97
6	107	Vijay	Pandas	72.0	68.0	75
7	108	Ananya	Python	88.0	85.0	91
8	109	Suresh	NumPy	NaN	77.0	81
9	110	Lakshmi	Pandas	93.0	96.0	94
10	111	Naveen	Python	58.0	64.0	61
11	112	Keerthi	NumPy	81.0	86.0	84
12	113	Manoj	Pandas	74.0	NaN	78
13	114	Sneha	Python	97.0	93.0	98
14	115	Ravi	NumPy	69.0	72.0	74

In [18]:
```
df["Total Marks"] = df["python_marks"] + df["numpy_marks"] + df["pandas_marks"]
```

In [19]:
```
df
```

Out[19]:

	student_id	name	course	python_marks	numpy_marks	pandas_marks	Total Marks
0	101	Arun	Python	85.0	78.0	92	255.0
1	102	Priya	Python	91.0	88.0	95	274.0
2	103	Karthik	NumPy	76.0	82.0	79	237.0
3	104	Divya	Pandas	89.0	94.0	90	273.0
4	105	Rahul	Python	67.0	73.0	70	210.0
5	106	Meena	NumPy	95.0	91.0	97	283.0
6	107	Vijay	Pandas	72.0	68.0	75	215.0
7	108	Ananya	Python	88.0	85.0	91	264.0
8	109	Suresh	NumPy	NaN	77.0	81	NaN
9	110	Lakshmi	Pandas	93.0	96.0	94	283.0
10	111	Naveen	Python	58.0	64.0	61	183.0
11	112	Keerthi	NumPy	81.0	86.0	84	251.0
12	113	Manoj	Pandas	74.0	NaN	78	NaN
13	114	Sneha	Python	97.0	93.0	98	288.0
14	115	Ravi	NumPy	69.0	72.0	74	215.0

In [25]:
```
df.groupby("course").count()
```

Out[25]:

course	student_id	name	python_marks	numpy_marks	pandas_marks	Total Marks
NumPy	5	5	4	5	5	4
Pandas	4	4	4	3	4	3
Python	6	6	6	6	6	6

In [27]:
```
array = np.array([1200,1400,1600,4343,343,4522])
np.sum(array)
```

Out[27]:
```
np.int64(13408)
```

In [28]:
```
np.mean(array)
```

Out[28]:
```
np.float64(2234.6666666666665)
```

In [29]:
```
np.min(array)
```

Out[29]:
```
np.int64(343)
```

In [30]:
```
np.max(array)
```

Out[30]: np.int64(4522)

In [ ]: